Year 11 Higher

REVISION PACK

Shape and Space 6

ANSWERS

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Shape and Space 6 · Angles in polygons (Lessons 10.2, 10.3, 10.4)

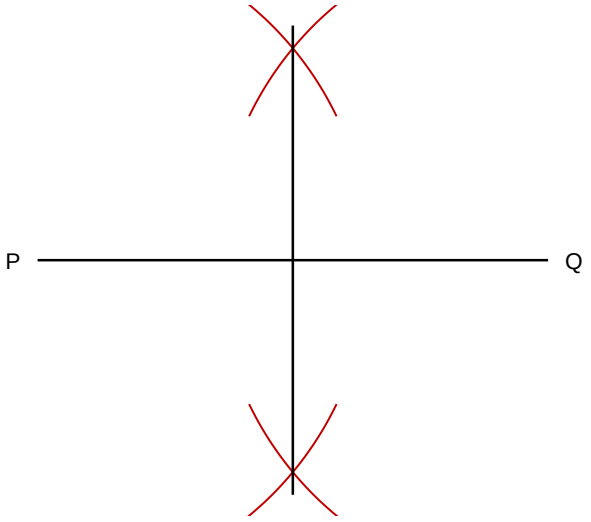
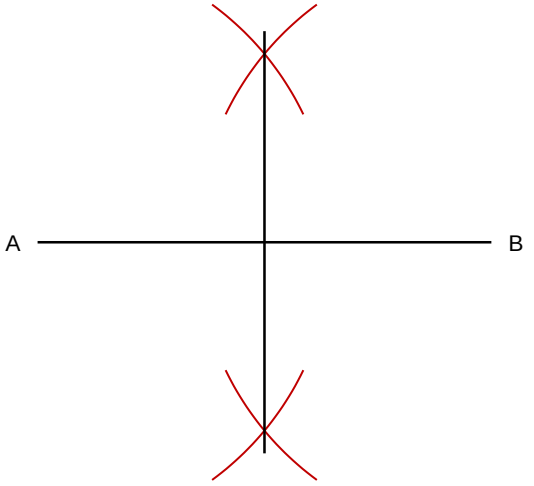
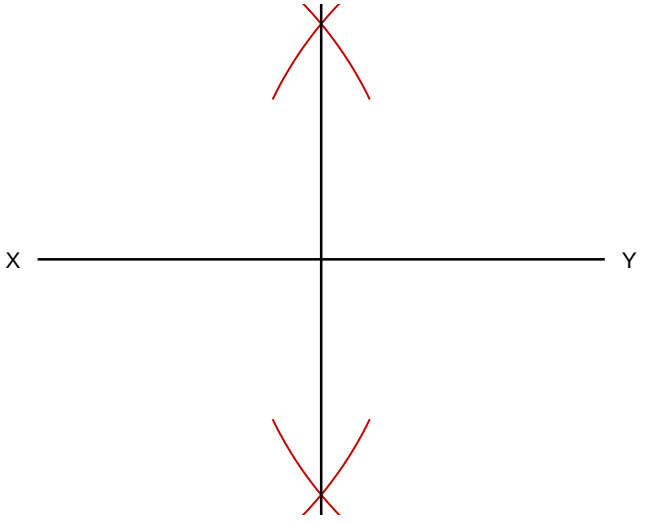
Q	Working	Answer
regular polygons		
1.1	$360 \div 20 = 18$	**18 sides**
1.2	$180 - 144 = 36$ $360 \div 36 = 10$	**10 sides**
1.3	$360 \div 36 = 10$	**10 sides**
1.4	$180 - 108 = 72$ $360 \div 72 = 5$	**5 sides**
1.5	$360 \div 45 = 8$	**8 sides**
1.6	$180 - 135 = 45$ $360 \div 45 = 8$	**8 sides**
finding x in a polygon		
1.7	$(6 - 2) \times 180 = 720$ $110 + 130 + 140 + 130 = 510$ $3x = 720 - 510 = 210$ $x = 70$	$x = 70$
1.8	$(5 - 2) \times 180 = 540$ $100 + 110 + 120 = 330$ $2x = 540 - 330 = 210$ $x = 105$	$x = 105$
1.9	$(7 - 2) \times 180 = 900$ $140 + 150 + 130 + 120 + 120 = 660$ $3x = 900 - 660 = 240$ $x = 80$	$x = 80$

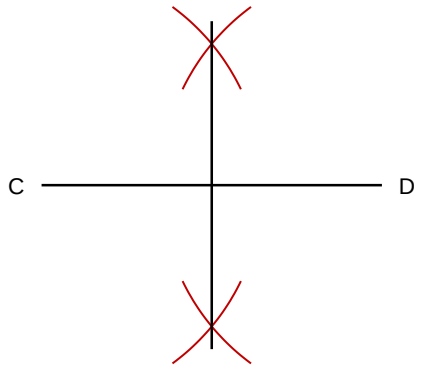
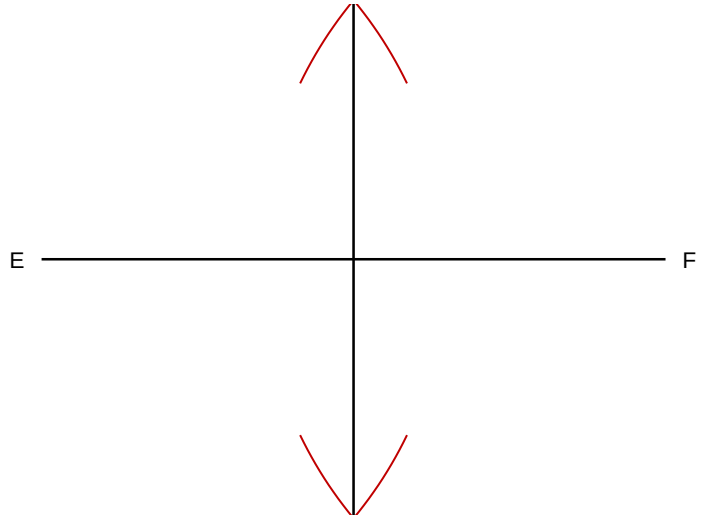
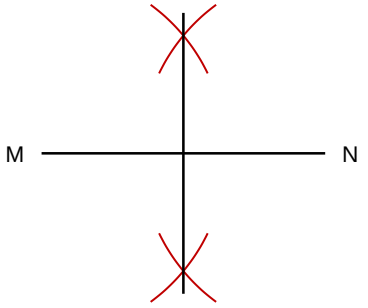
1.10	$(5-2)\times 180 = 540$ $100+140+120 = 360$ $2x = 540-360 = 180$ $x = 90$	$x = 90$
1.11	$(6-2)\times 180 = 720$ $120+125+135+145 = 525$ $3x = 720-525 = 195$ $x = 65$	$x = 65$
1.12	$(5-2)\times 180 = 540$ $90+110+120 = 320$ $4x = 540-320 = 220$ $x = 55$	$x = 55$
angles in a kite		
1.13	$360-64-50 = 246$ $246 \div 2 = 123$	**123°**
1.14	$360-100-70 = 190$ $190 \div 2 = 95$	**95°**
1.15	$360-90-40 = 230$ $230 \div 2 = 115$	**115°**
1.16	$360-56-80 = 224$ $224 \div 2 = 112$	**112°**
1.17	$360-86-40 = 234$ $234 \div 2 = 117$	**117°**
1.18	$360-110-34 = 216$ $216 \div 2 = 108$	**108°**

Shape and Space 6 · Congruence and similarity (Lessons 10.1, 10.6)

Q	Working	Answer
proving triangles are congruent		

2.1	$AB = AD$ (given) $CB = CD$ (given) AC is common	**congruent (SSS)**
2.2	$AB = CD$ (opposite sides) $AD = CB$ (opposite sides) BD is common	**congruent (SSS)**
2.3	$AB = CD$ (opposite sides) $BC = DA$ (opposite sides) AC is common	**congruent (SSS)**
2.4	$PX = SX$ (given) $QX = RX$ (given) $\angle PXQ = \angle SXR$ (vertically opposite)	**congruent (SAS)**
similar triangles with parallel lines		
2.5	$(a)\ 2 + 6 = 8$ $8 \div 2 = 4$ $3 \times 4 = 12$ $(b)\ 1.5 \times 4 = 6$ $6 - 1.5 = 4.5$	**(a) 12 cm** **(b) 4.5 cm**
2.6	$(a)\ 3 + 9 = 12$ $12 \div 3 = 4$ $4 \times 4 = 16$ $(b)\ 2.5 \times 4 = 10$ $10 - 2.5 = 7.5$	**(a) 16 cm** **(b) 7.5 cm**
2.7	$(a)\ 5 + 3 = 8$ $8 \div 5 = 1.6$ $7.5 \times 1.6 = 12$ $(b)\ 4 \times 1.6 = 6.4$ $6.4 - 4 = 2.4$	**(a) 12 cm** **(b) 2.4 cm**
2.8	$(a)\ 2 + 3 = 5$ $5 \div 2 = 2.5$ $4 \times 2.5 = 10$ $(b)\ 1.8 \times 2.5 = 4.5$ $4.5 - 1.8 = 2.7$	**(a) 10 cm** **(b) 2.7 cm**

the perpendicular bisector		
2.9		**Equal arcs from P and Q above and below; the line crosses PQ at 4.5 cm from each end at 90°**
2.10		**Equal arcs from A and B above and below; the line crosses AB at 4 cm from each end at 90°**
2.11		**Equal arcs from X and Y above and below; the line crosses XY at 5 cm from each end at 90°**

2.12		**Equal arcs from C and D above and below; the line crosses CD at 3 cm from each end at 90°**
2.13		**Equal arcs from E and F above and below; the line crosses EF at 5.5 cm from each end at 90°**
2.14		**Equal arcs from M and N above and below; the line crosses MN at 2.5 cm from each end at 90°**

Shape and Space 6 · Constructions

Q	Working	Answer
the bisector of an angle		
8.1	Arcs from Q and from both arms shown Check with a protractor	**Each half is 35°**
8.2	Arcs from Q and from both arms shown Check with a protractor	**Each half is 55°**
8.3	Arcs from Q and from both arms shown Check with a protractor	**Each half is 22.5°**
8.4	Arcs from Q and from both arms shown Check with a protractor	**Each half is 65°**